

Introducing think▶then

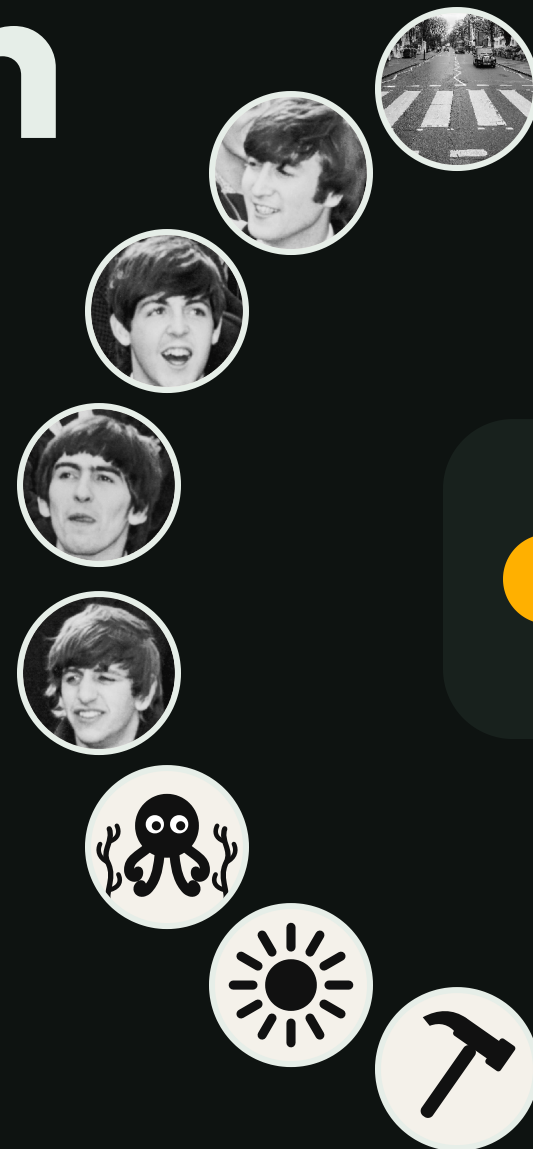
October 1, 2026



Ian Maurer



GenomOncology



decide

"Hey Jude" · written for Julian?
true [0.92]

choose

"Here Comes the Sun" · singer?
George [0.99]

tag

"Blackbird" · which describe its music?
acoustic [0.92] · **melancholy** [0.58]

score

"Yesterday" · how long, 0 to 9 minutes?
2.12

filter

12 songs · on Abbey Road?
Octopus's Garden [0.75] ...

rank

30 songs · most covered Beatles song?
Yesterday [0.89]

find

70 songs · about John's mother?
Julia [0.78]

annotate

"Yesterday"
mood: nostalgic · **instrument: strings** · **singer: Paul ...**

recognize

George Harrison PERSON [1.00] · **Something** SONG [1.00]

relate

Ringo Starr sings → **Octopus's Garden** [0.94]
John Lennon sings → **Come Together** [0.84] ...



Code sees strings, not meaning.

CODE SEES

"Abbey Road"

str

"Octopus's Garden"

str

"Ringo Starr"

str

YOU SEE



Abbey Road
ALBUM

↑ appears on

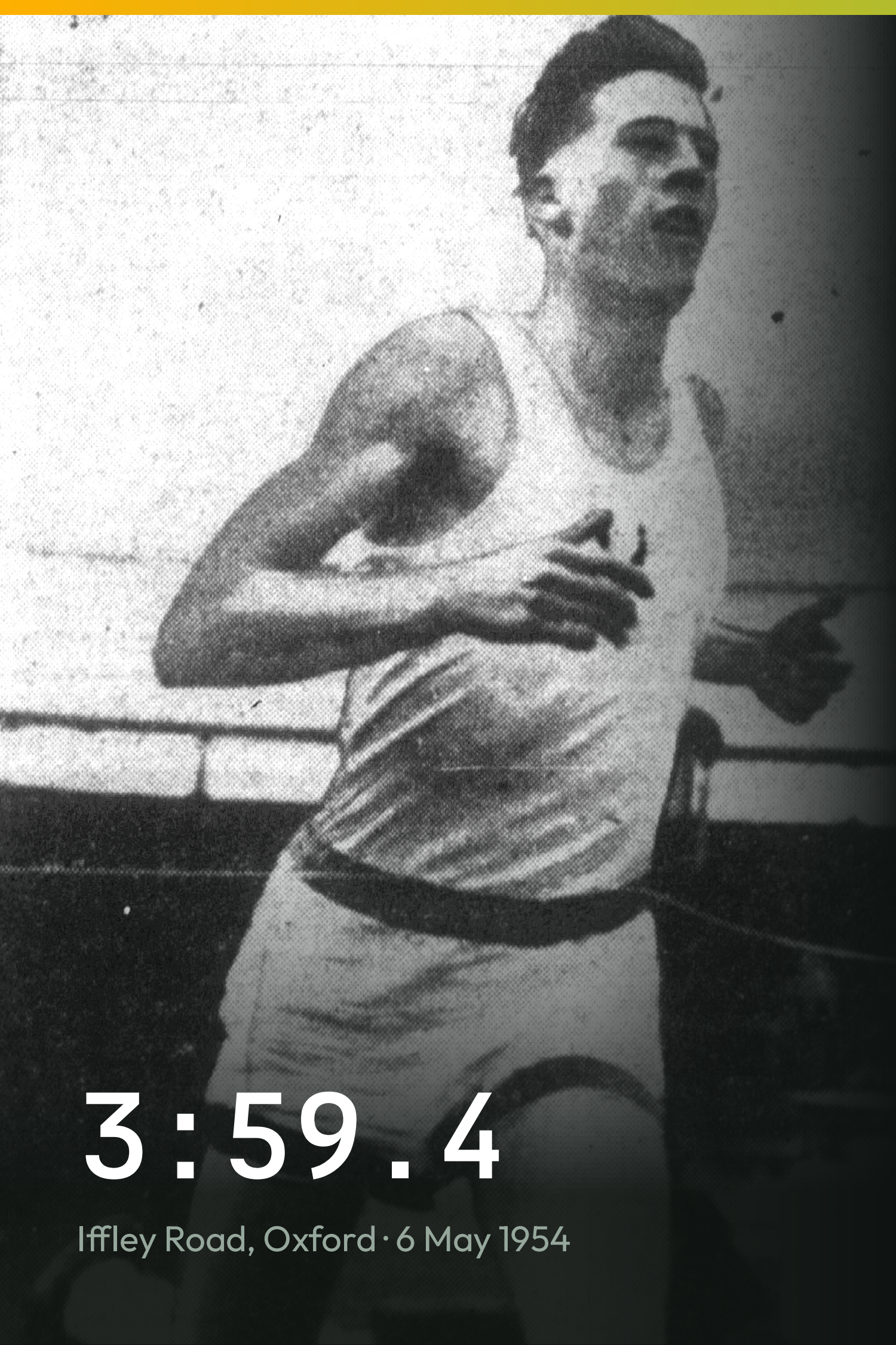


Octopus's Garden
SONG

↑ sings



Ringo Starr
DRUMMER



Jev set the standard.

text

"Octopus's Garden"

question

who is the lead singer?

60 ms

choose

\$0.000016



John

0.01



Paul

0.02



George

0.13



Ringo

0.84



duet

0.00

3:59.4

Iffley Road, Oxford · 6 May 1954

- fast
- cheap
- hosted
- easy to use
- no training

Jev predicts. It never generates text.

Beatles Bench is our playground.

SONG	RELEASED	FIRST ALBUM	WRITTEN BY	LEAD	LENGTH	VIEWS 2024
Love Me Do	Oct 1962	Please Please Me	Lennon-McCartney	McCartney and Lennon	2:19	153,789
She Loves You	Aug 1963	Past Masters	Lennon-McCartney	Lennon and McCartney	2:21	103,629
Yesterday	Aug 1965	Help!	Lennon-McCartney	McCartney	2:05	239,052
Octopus's Garden	Sep 1969	Abbey Road	Starkey	Starr	2:51	103,246
Here Comes the Sun	Sep 1969	Abbey Road	Harrison	Harrison	3:05	272,409
Hey Jude	Aug 1968	Past Masters	Lennon-McCartney	McCartney	7:08	477,821

306 songs · 1,501 questions
github.com/botassembly/beatles-bench

10 functions, 1 CLI, 24 bindings.

```
$ cat songs.txt | thinkthen filter --lines --threshold 0.78 \  
    "Is this song on the album Abbey Road?"  
Come Together  
Something  
...
```

FUNCTIONS

decide

filter

rank

choose

find

score

tag

annotate

recognize

relate

TOOLS

audit

diff

check

cache

status

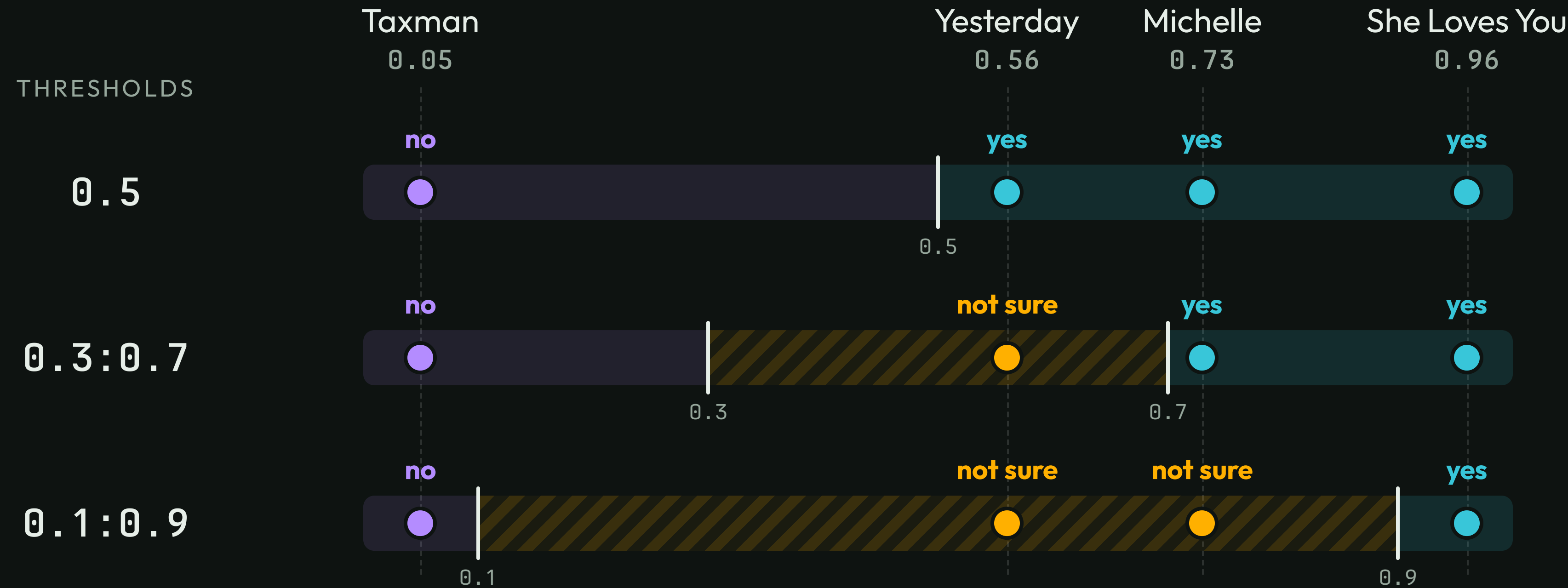
transform

OPTIONS

- threshold 0.78 the bar
- threshold 0.3:0.7 not-sure band
- lines --jsonl --csv input format
- field /title one field
- details full answer
- cache DIR reuse answers
- url URL custom backend
- jobs 8 in parallel

-  CLI
-  Ada
-  C
-  C#
-  C++
-  COBOL
-  Dart
-  DuckDB
-  Go
-  Java
-  Kotlin
-  Objective-C
-  pandas
-  PHP
-  Polars
-  PostgreSQL
-  Python
-  R
-  Ruby
-  Rust
-  Scala
-  SQLite
-  Swift
-  TypeScript
-  Zig

decide answers yes, no, or not sure.



“Is this a love song?”

choose selects one option.

REAL ANSWER BAR 0.5



John



Paul



George



Ringo

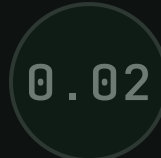


John + Paul

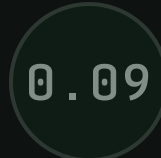
Come Together



Yesterday



Something



Octopus's Garden

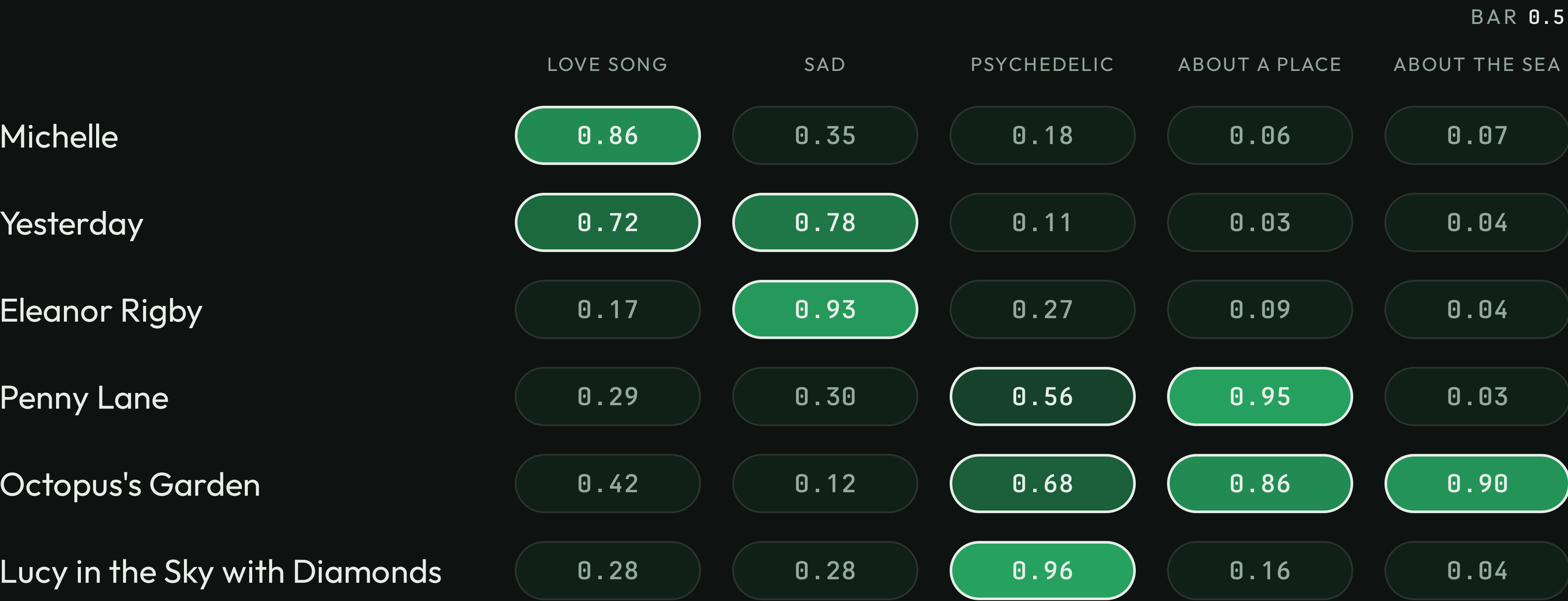


She Loves You



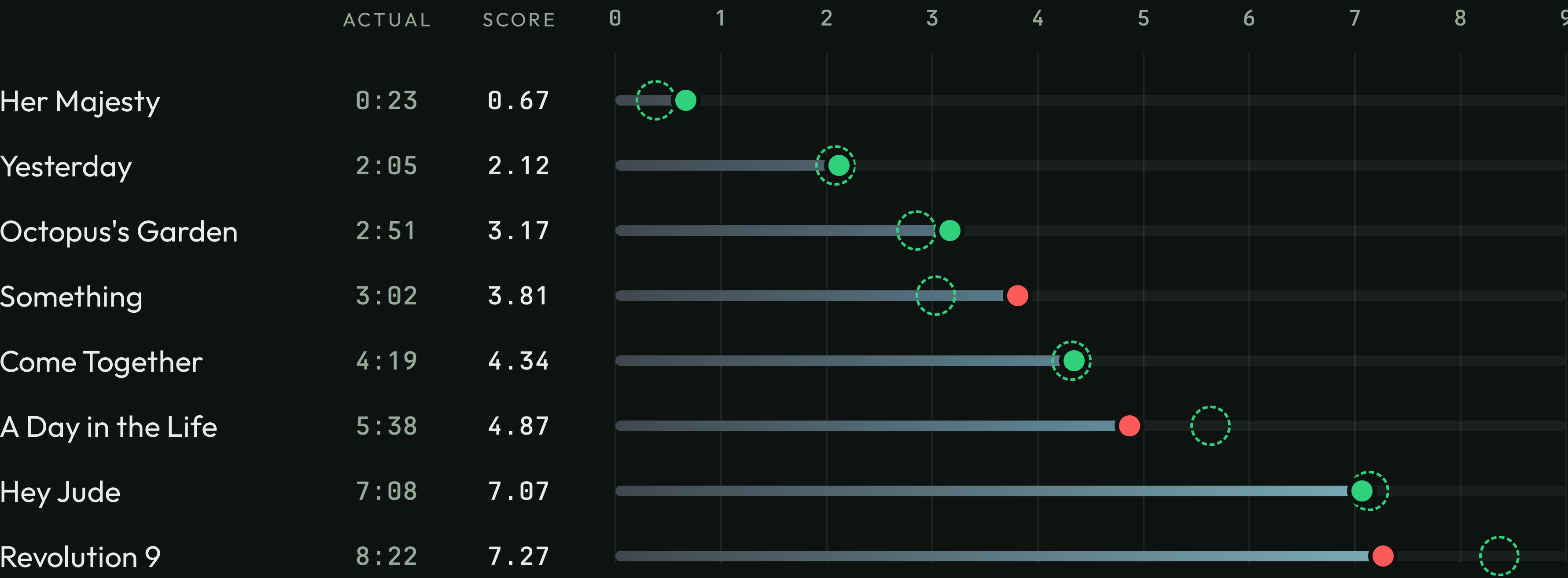
“Who sings the lead vocal?”

tag applies labels to data.



“Which of these describe it?”

score can rate on a linear scale you define.



“How long is the recording, in minutes?”

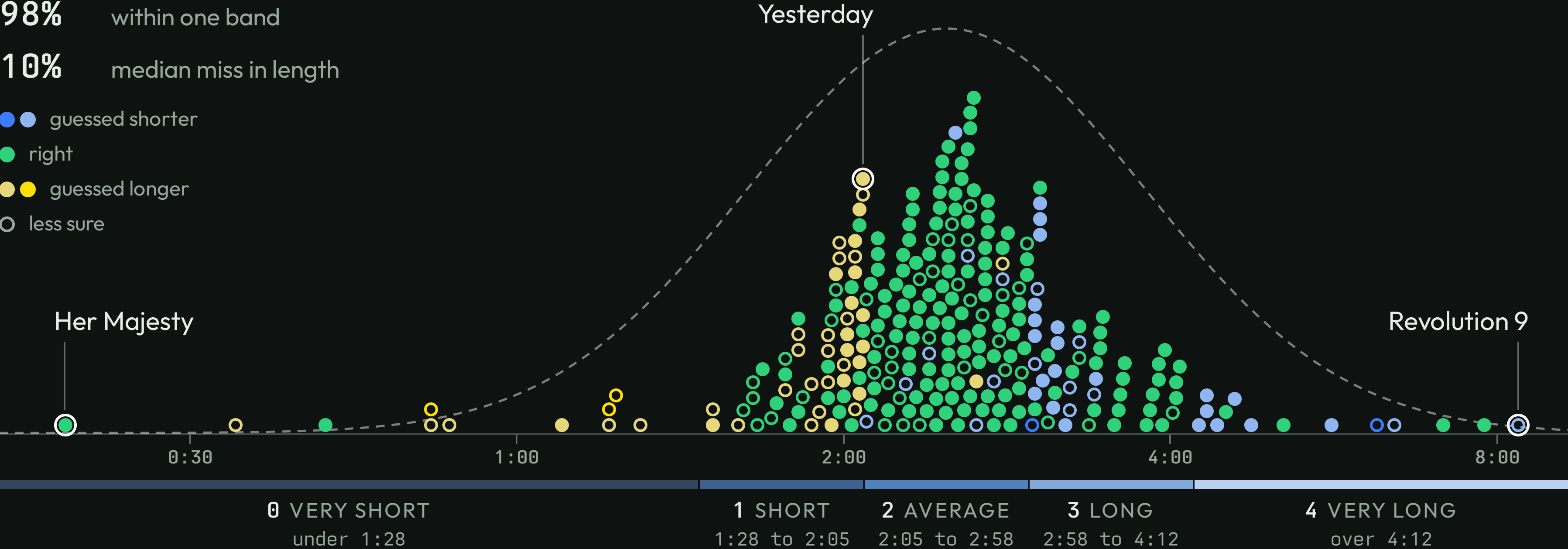
score can rate on a non-linear scale you define.

70% in the right band

98% within one band

10% median miss in length

- guessed shorter
- right
- guessed longer
- less sure



“How long is the recording?”

filter keeps what clears your bar.

IN, AS THEY CAME

Octopus's Garden		0.75
Yellow Submarine		0.12
Something		0.80
Here Comes the Sun		0.96
Yesterday		0.11
Come Together		0.95
Hey Jude		0.29
Penny Lane		0.43
A Day in the Life		0.93
Her Majesty		0.90
Let It Be		0.27
Maxwell's Silver Hammer		0.94

BAR
0.7
→

OUT, SAME ORDER

0.75	Octopus's Garden	✓
0.80	Something	✓
0.96	Here Comes the Sun	✓
0.95	Come Together	✓
0.93	A Day in the Life	✗
0.90	Her Majesty	✓
0.94	Maxwell's Silver Hammer	✓

“Is it on Abbey Road?”

rank sorts by your criteria.

BAR 0.8



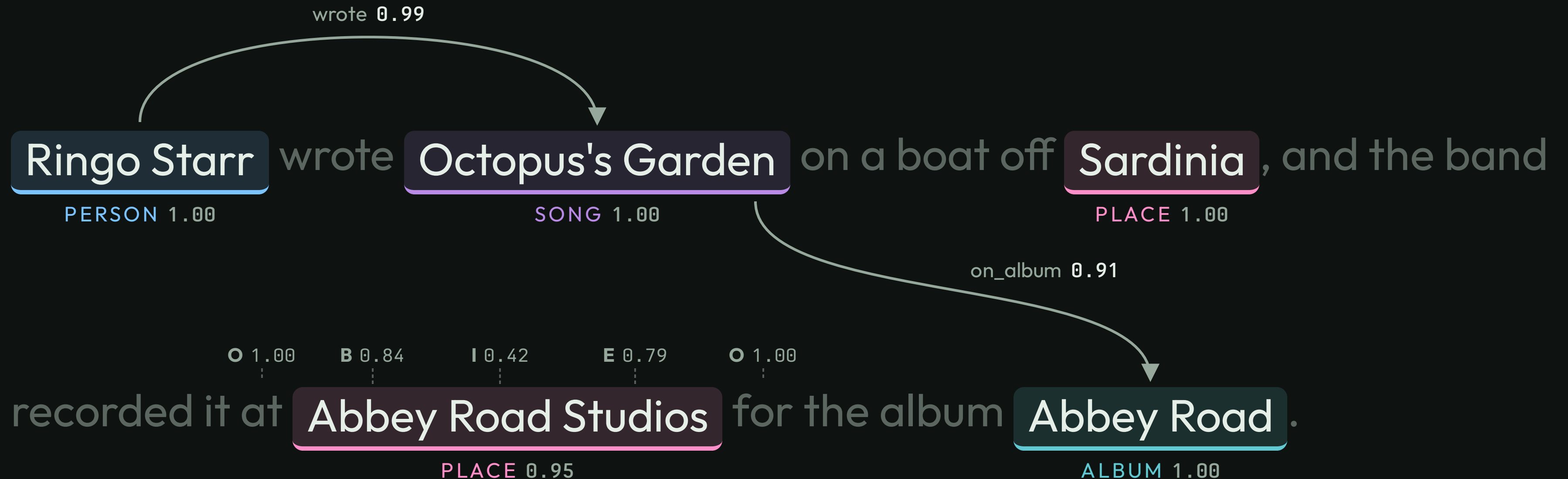
“Is it one of the Beatles' biggest hits?”

find picks one from many.

She Loves You 0.01	I Saw Her Standing There 0.01	From Me to You 0.08	All My Loving 0.00	Love Me Do 0.63 ✓
I Want to Hold Your Hand 0.00	Please Please Me 0.27	Can't Buy Me Love 0.00	A Hard Day's Night 0.00	I Feel Fine 0.00

“Which one did they release first?”

recognize labels things it finds.



“Which names are people, songs, albums, or places?”

relate asks what Jev knows about each pair.

NAMES IN		THE COMMAND		EDGES OUT, 0.5 OR MORE		
John Lennon	PERSON	→ thinkthen relate @rules.json PREDICATE SUBJECT OBJECT sings = person → song on_album = song → album →		sings		
Paul McCartney	PERSON			John Lennon	→ Come Together	0.84 ✓
George Harrison	PERSON			Paul McCartney	→ Yesterday	0.77 ✓
Ringo Starr	PERSON			George Harrison	→ Something	0.91 ✓
				Ringo Starr	→ Octopus's Garden	0.94 ✓
Come Together	SONG			on_album		
Yesterday	SONG			Come Together	→ Abbey Road	0.97 ✓
Something	SONG			Yesterday	→ Help!	0.65 ✓
Taxman	SONG			Yesterday	→ Revolver	0.64 ✗
Octopus's Garden	SONG			Something	→ Revolver	0.62 ✗
				Something	→ Abbey Road	0.93 ✓
Help!	ALBUM			Taxman	→ Revolver	0.95 ✓
Revolver	ALBUM			Octopus's Garden	→ Revolver	0.90 ✗
Abbey Road	ALBUM			Octopus's Garden	→ Abbey Road	0.63 ✓

Store and Reuse Questions

original.json

```
#!/ JSON
{
  "decide": "Did a Beatle write this song?",
  "true": "John, Paul, George, or Ringo wrote it.",
  "false": "Someone else wrote it. It is a cover.",
  "threshold": "0.2:0.8",
  "on": ["/title", "/album"]
}
```

run

```
#!/ BASH
echo '{"title":"Something","album":"Abbey Road"}' |
  thinkthen decide @original.json
code=$?
echo "exit $code"
```

output

```
#!/ TEXT
true
exit 0
```

annotate answers many questions at once.

questions.json

```
#!/ JSON
{
  "version": 1,
  "questions": {
    "singer": {
      "choose": "Who sings the lead vocal?",
      "options": ["John", "Paul", "George", "Ringo",
                  "John and Paul duet"]
    },
    "tags": {
      "tag": "Which of these describe it?",
      "labels": ["love song", "sad", "psychedelic",
                  "about a place", "about the sea"]
    },
    "abbey_road": {"decide": "Is it on Abbey Road?"},
    "big_hit": {"decide": "Is it one of the Beatles' biggest hits?"}
  }
}
```

run

```
#!/ BASH
$ cat songs.jsonl
{"title": "Yesterday"}
{"title": "Octopus's Garden"}
$ thinkthen annotate questions.json \
  --jsonl --field /title < songs.jsonl
```

output

```
#!/ JSON
{"title": "Yesterday",
 "singer": "Paul",
 "tags": ["love song", "sad"],
 "abbey_road": false, "big_hit": true}

{"title": "Octopus's Garden",
 "singer": "Ringo",
 "tags": ["psychedelic", "about a place",
          "about the sea"],
 "abbey_road": true, "big_hit": false}
```

Bash runs on judgment.

setlist.sh

```
#!/ BASH
#!/usr/bin/env bash
is_original() {
    thinkthen decide 'Did a Beatle write this song?' \
        --true 'John, Paul, George, or Ringo wrote it.' \
        --false 'Someone else wrote it. It is a cover.' \
        --threshold 0.2:0.8 --quiet \
        --field /title --field /album
}
while read -r song; do
    title=$(printf '%s\n' "$song" | jq -r .title)
    printf '%s\n' "$song" | is_original
    code=$?
    case $code in
        0) echo "play    $title" ;;
        1) echo "skip    $title" ;;
        3) echo "check   $title" ;;
        *) exit "$code" ;;
    esac
done
```

setlist.jsonl

```
#!/ JSON
{"title":"Something","album":"Abbey Road"}
{"title":"Act Naturally","album":"Help!"}
{"title":"Words of Love","album":"Beatles for Sale"}
{"title":"Mr. Moonlight","album":"Beatles for Sale"}
```

run

```
#!/ BASH
$ ./setlist.sh < setlist.jsonl
play    Something          # 0 (exit code)
skip    Act Naturally      # 1
play    Words of Love      # 0
check   Mr. Moonlight      # 3
```

Scripting and systems languages

```
#!/ PYTHON

import thinkthen as tt

question = (
    "The text is the title of a song by the Beatles. "
    "Ringo Starr sings the lead vocal on it."
)

song = "Octopus's Garden"

is_ringo = tt.decide(
    question,
    song,
).value

assert is_ringo is True
```

```
#!/ C

thinkthen_engine *tt = thinkthen_engine_new();
const char *question =
    "The text is the title of a song by the Beatles. "
    "Ringo Starr sings the lead vocal on it.";
const char *song = "Octopus's Garden";

thinkthen_answer is_ringo;
int status = thinkthen_decide(
    tt,
    question,
    song,
    strlen(song),
    &is_ringo);

assert(status == THINKTHEN_OK);
assert(is_ringo.outcome == THINKTHEN_YES);
```

 Python

 TypeScript

 Ruby

 PHP

 Bash

 R

 C

 Go

 Java

 Ada

 C++

 Zig

 Kotlin

 COBOL

 C#

 Swift

 Scala

 Rust

 Objective-C

 Dart

Database tables and data frames

```
#! SQL
SELECT title, thinkthen_decide('{
  "decide": "The text is the title of a song by the Beatles. It appears on the album Abbey Road.",
  "threshold": "0.3:0.7"
}', title) AS on_abbey
FROM songs;
```

title	on_abbey
Here Comes the Sun	✓
Yellow Submarine	✗
Octopus's Garden	✓
Penny Lane	?
A Day in the Life	✓
Hey Jude	?

 **SQLite**

 **DuckDB**

 **PostgreSQL**

 **Polars**

 **pandas**

 **R**

Jev has blind spots.

Obscure songs



Who sings lead on Nothin' Shakin'?

- ✗ Ringo Starr 0.34
- ✓ George Harrison 0.17

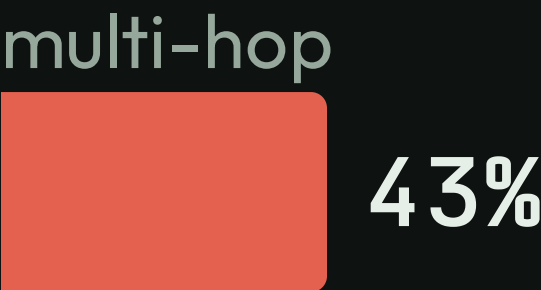
Tricky wording



Which song first came out on the album Yellow Submarine?

- ✗ Yellow Submarine 0.52
- ✓ It's All Too Much 0.34

Multi-hop



Did Can't Buy Me Love come out the same month as the 1964 Alaska earthquake?

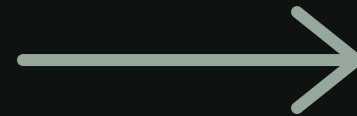
- ✗ No 0.65
- ✓ Yes 0.35

Retrieval-augmented decisions (RAD)

Giving Jev context (i.e., state) improves accuracy.

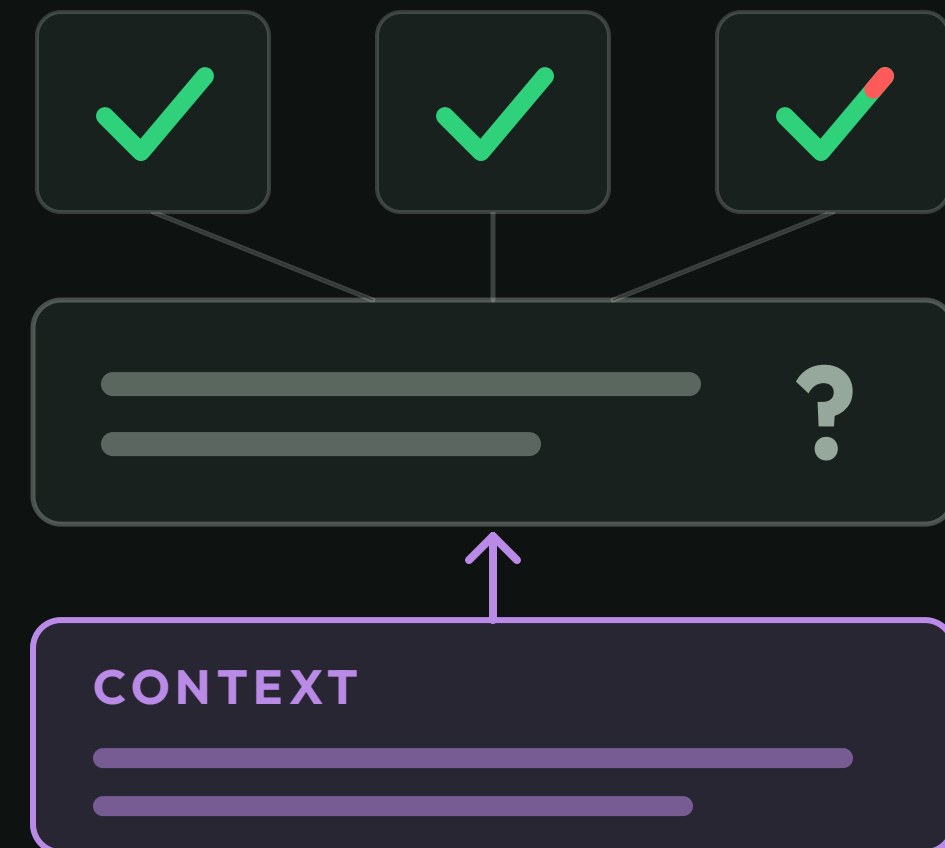
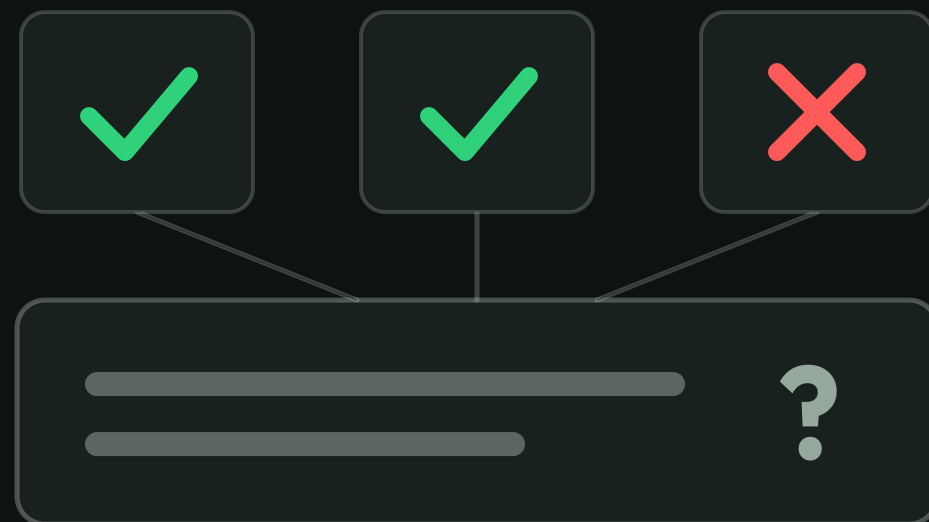
NO CONTEXT

68%



WITH CONTEXT

97%



audit grades a run.

SONG	ANSWER	NO CONTEXT
A Day in the Life	no	0.97 ✖
Blackbird	no	0.32 ?
Come Together	yes	0.96 ✔
Glass Onion	no	0.62 ?
Here Comes the Sun	yes	0.97 ✔
In My Life	no	0.59 ?
Octopus's Garden	yes	0.80 ✔
Something	yes	0.78 ?
Taxman	no	0.77 ?
The Long and Winding Road	no	0.92 ✖

“Is it on Abbey Road?”

THRESHOLD 0.2:0.8

diff shows what changed.

SONG	ANSWER	NO CONTEXT		WITH CONTEXT
A Day in the Life	no	0.97 ✖	→	0.12 ✔
Blackbird	no	0.32 ?	→	0.02 ✔
Glass Onion	no	0.62 ?	→	0.02 ✔
In My Life	no	0.59 ?	→	0.06 ✔
Something	yes	0.78 ?	→	0.98 ✔
Taxman	no	0.77 ?	→	0.03 ✔
The Long and Winding Road	no	0.92 ✖	→	0.18 ✔

“Is it on Abbey Road?”

THRESHOLD 0.2:0.8

Agent harnesses

Permission

“Should this command run?”

If not sure, ask a person.

decide

Tool choice

“Which tools fit this step?”

Load only the top few.

rank

Context

“What should happen to this chunk?”

Load, summarize or hide it.

choose

Model choice

“Is this step easy?”

Send it to a smaller model.

decide

Parallel work

“Can these tasks run at once?”

Split them.

decide

Data safety

“Could this touch secrets?”

Use an approved model.

decide

Off track

“Is the agent going in circles?”

Stop it and ask a person.

decide

Done check

“Is the task done as asked?”

Stop, or send it back.

decide

Evaluation

“Which answer is better?”

Pick the winning prompt.

choose

Ops/Security

Alert triage

“Does this alert need a person now?”

Page, log, or queue it.

`decide`

Failure routing

“What kind of failure is this?”

Run its runbook.

`choose`

Grep by meaning

“Is this line about the outage?”

Keep it, whatever words it uses.

`filter`

Login check

“How suspicious is this session?”

Send the top two levels to a person.

`score`

Flaky test

“Is this failure flaky?”

Retry, or fail the build.

`decide`

Release notes

“Feature, fix or breaking change?”

Label each commit. Bump the version.

`tag`

Sample sheet

“Is this row complete?”

Stop before hours of compute.

`filter`

Incident email

“What service, severity and start time?”

Hand the JSON to jq.

`annotate`

Worst first

“Which error hurts users most?”

Fix from the top.

`rank`

Data science

Filter by meaning

“Is this a complaint?”

Keep the yes rows.

`filter`

Category column

“Which category is this?”

Group by the new column.

`choose`

Extract fields

“What amount, date and vendor?”

Free text becomes columns.

`annotate`

Find duplicates

“Are these the same company?”

Merge the matching pairs.

`decide`

Score and average

“How happy is this review?”

Average by product.

`score`

Count topics

“Which topics does this ticket cover?”

Unnest and count them.

`tag`

Check at load

“Is this address complete?”

Send bad rows to quarantine.

`decide`

Top N per group

“Is this lead ready to buy?”

Keep the top 3 per account.

`rank`

Second opinion

“Does the label fit this row?”

List where model and label disagree.

`decide`

Four ways to retrieve.

Keyword

Matches shared words

TF-IDF, BM25

Semantic

Matches similar meaning

Embeddings, cosine similarity

Hybrid

Blends both scores

BM25 plus embeddings

Classification

Answers a question about each item

Language model

Combine them: search narrows the set, and classification judges what is left.

Two weeks later, the field arrived.

RELEASED

Liquid d1

The first decision foundation model, from an MIT spin-off.

Speaks the System One API. Launched Sep 29.

SERVED LOCALLY

Ollama 0.35

nimble and tevl speak the System One API.

Shipped Sep 30.

OPEN WEIGHTS

~~62~~ 106 models on JevBench

Cygnet tops the composite, with Jev third.

As of Sep 30.

ANNOUNCED

OpenAI Decisions API

Luna answers predefined questions over text or images.

Announced at DevDay, Sep 29.

Think Then: one SDK for every decision API.

Beatles Bench scores the field.

SYSTEM	WHERE	TOTAL	HARD	EASY
Jev	hosted	70.5	56.9	77.3
Liquid d1	hosted	67.6	52.9	75.0
Nimble 9B	local	51.9	41.6	57.1
Kev-4B	local	47.2	39.0	51.4
Laya	local	35.8	29.9	38.8
GLM-5.3 Flash no reasoning	hosted	96.7	95.4	97.3

Bring your own backend.

```
$ thinkthen check --url http://127.0.0.1:39519/arm/full/v1
url http://127.0.0.1:39519/arm/full/v1/systemone
model asked unspecified
model sent jev-1.13.0
ok connection
ok key
ok endpoint
ok noul
ok choice
ok score
ok mixed
ok usage
critical 0, warning 0
```

Open an issue: <https://github.com/botassembly/awesome-thinkthen>

BioMCP A biomedical search tool for agents

1 command, 15 entities.

```
$ biomcp search <entity> [query|filters]
$ biomcp get <entity> <id> [section...]
$ biomcp get variant "BRAF V600E"
```

ENTITIES

gene	variant	article	author	trial
diagnostic	drug	disease	phenotype	pgx
gwas	pathway	cell-line	protein	adverse-event

70 SOURCES

PubMed

ClinicalTrials.gov

ClinVar

gnomAD

OncoKB

CIViC

cBioPortal

Reactome

UniProt

OpenFDA

...and 60 more

biomcp.org

github.com/genomoncology/biomcp

\$  | botassembly

Understandable Markdown Agents.

THE TRIAGE EXAMPLE

```
triage/  
  ASSEMBLY.md  
  flows/triage/  
    FLOW.md  
    01-classify/  
      STAGE.md  
      schema.json  
      before  
    02-route/  
      CHOOSE.md  
    03-verify/  
      STAGE.md  
    gate/  
  skills/priority-rubric/SKILL.md
```

THE PRINCIPLES

Understandable

Plain English a person can read without the runtime.

Durable

The files outlive the program, review as a diff, live in git.

Constrained, not hoped

Scripts and gates fence the agent on every side.

A folder in, a record out

The whole job. Everything else is vocabulary.

Progressive disclosure

Told what exists by name. It reads the rest when it reaches.

github.com/botassembly/botassembly

Pathology Workbench

Quality control

Review

Sign-off

Labs send signed
reports to the clinician
and the health record.

Precision Oncology Platform

Therapies

Clinical trials

Annotations

Curated knowledge
drives trial matching
and decision support.

Biomedical Agents

BioMCP

Bot Assembly

ThinkThen

Agents cite their
evidence, check each
step, and record each run.

think▶then

code that knows what you mean

`thinkthen.dev`

`github.com/botassembly/thinkthen`



Ian Maurer



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